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# LINEAR PROGRAMMING WITH MULTIPLE OBJECTIVE FUNCTIONS: STEP METHOD (STEM) *

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This paper describes a solution technique for Linear Programming problems with multiple objective functions. In this type of problem it is often necessary to replace the concept of "optimum" with that of "best compromise". In contrast with methods dealing with a priori weighted sums of the objective functions, the method described here involves a sequential exploration of solutions. This exploration is guided to some extent by the decision maker who intervenes by means of defined responses to precise questions posed by the algorithm. Thus, in this man-model symbiosis, phases of computation alternate with phases of decision. The process allows the decision-maker to "learn" to recognize good solutions and the relative importance of the objectives. The final decision (best compromise) furnished by the man-model system is obtained after a small number of successive phases.

## 1. Introduction

A large number of real problems can be transformed into LP models. However, some types of problems cannot be exactly transformed in this manner, to give a unique objective function unless extra "arbitrary" constraints are added. We will identify by MOLP models (Multi-Objective Linear Programming) those LP models requiring several, simultaneous objective functions (or more briefly "objectives"). Such models have the advantage of accurately representing the real multicriteria

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nature of certain situations. Among the problems which can be represented by MOLP, we can list the following ones:

- production: maximisation of benefits, minimisation of manufacturing time of the products, minimisation of the number of defective parts produced [4];
- transportation: minimisation of transport costs, maximisation of delivery speed, maximisation of goods delivered [7];
- scheduling: minimisation of delay, minimisation of total cost, optimum allocation of resources;
- personnel allocation: minimisation of total cost (salaries, recruitement, training), maximisation of efficiency of personnel, minimisation of "slack" between forecast staff requirements and programme allocations [2].

In the above examples the objectives sought are often conflicting, sometimes even opposing, in all cases the relative "weight" of each is difficult to assess a priori.

Consider the mathematical formulation of a typical MOLP problem. Let the objective functions be:

$$\begin{aligned} C^1 &= \sum_{i=1}^{n} c_i^1 \cdot x_i \\ &\cdots\cdots \\ C^j &= \sum_{i=1}^{n} c_i^j \cdot x_i \\ &\cdots\cdots \\ C^k &= \sum_{i=1}^{n} c_i^k \cdot x_i \,. \end{aligned} \tag{1}$$

We want to maximise the objective functions *, i.e. we want to obtain values as high as possible for each of the $C^j$ under the constraints.

$$\begin{aligned} A \cdot X &\leq b \qquad X = (x_1, x_2, \ldots, x_n)' \\ x_i &\geq 0 \qquad i = 1, 2, \ldots, n \,. \end{aligned} \tag{2}$$

* When an objective function is to be minimised, it is assumed to be transformed to maximisation by an appropriate sign change.

Let $D$ be the feasible region defined by constraints (2) with the matrix $A$ $(p \times n)$ and the vector $b(p \times 1)$. In general there are no feasible solutions such that all the $C^j$ can simultaneously take their maximum value within the feasible region $D$. Let us define by $M^j$ the optimum value of $C^j$ within $D$. The MOLP models will be characterised by:

$$\nexists \tilde{X} \in D \text{ such that } C^j(\tilde{X}) = M^j, \; j = 1, 2, ..., k . \tag{3}$$

The problem is to find the solution $X$ which will be the "best compromise" between the conflicting objectives.

The concept of "best compromise" is not yet well-defined. Some methods attempt to give a precise a priori definition of what is the best compromise. This can be done by defining a hierarchy (the objective functions are strictly ordered; the feasible region is restricted to solutions optimising the first objective function, then the second and so on) or by methods by "utility" (each objective functions $C^j$ is given a weight $p^j$, then a unique objective function $C = \Sigma_j C^j p^j$ is maximised). Other methods are heuristic: the exploration of a feasible region is conducted step by step toward the solution that will be finally chosen. Such methods are: parameterisation (the most important of the $C^j$ is maximised; the other $C^j$ are transformed into constraints, and the right and side is parameterised), or STEM.

## 2. The STEM method

### 2.1. *Essentials of STEM*

If the decision-maker is unable to give enough information about the relative importance of the objective functions, methods by hierarchy or utility are not accurate. Let us assume in this paper that we are in the extreme case: the decision maker does not have any a priori knowledge of the relative importance of the objective functions. *

STEM is an iterative exploration procedure, where the best compromise is reached after a certain number of cycles. Each cycle $m$ is made up of a calculation phase, and a decision-making phase, i.e. a conversation between the analyst and the decision-maker. During the decision-making phase, the decision-maker examines the results of the calcula-

* If he has some, the method can easily be modified to take into account this knowledge.

tion phase and from this examination he becomes able to give new information about his objectives.

2.2. *Construction of a Pay-off table T*

This table is constructed before the first cycle. For the feasible region $D$, defined by constraint (2), we calculate the optimum for each objective in turn and construct the Pay-off table $T$:

| | $C^1$ | $C^2$ | ... | $C^j$ | ... | $C^k$ |
|---|---|---|---|---|---|---|
| $C^1$ | | | | $z_j^1$ | | |
| $C^2$ | | | | $z_j^2$ | | |
| ⋮ | | | | ⋮ | | |
| $C^j$ | $z_1^j$ | $z_2^j$ | ... | $M^j$ | | $z_k^j$ |
| ⋮ | | | | ⋮ | | |
| $C^k$ | | | | $z_j^k$ | | |

(4)

Row $j$ corresponds to the solution vector $X^j$ maximising the objective function $C^j$, under the constraints (2). So $Z_i^j$ is the value taken on by the objective function $C^i$ when $C^j$ reaches its maximum $M^j$.

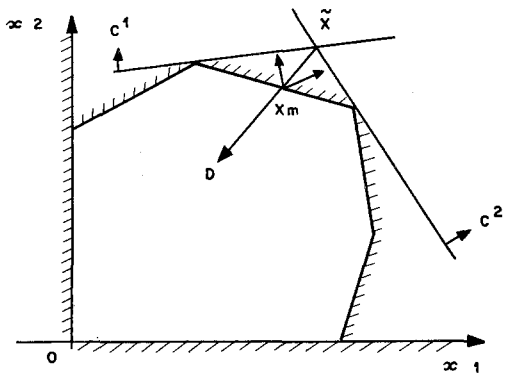
Let us denote as the "ideal solution" $\tilde{X}$ the point (which is not feasible according to property (3)) at which the various objective functions $C^j$ would take on the values $M^j$ exhibited on the main diagonal of the table (let $\tilde{Z}$ be the $k$-vector of the $M^j$ values).

2.3. *Calculation phase* (fig. 1)

For each cycle $m$, we will seek the feasible solution $X^m$ which is the "nearest", in the MINIMAX sense, to the ideal solution $\tilde{X}$, i.e. the solution * of the LP:

$$LP(m)\begin{cases}\min \lambda \\ \lambda \geqslant M^j - C^j(X) \cdot \pi^j \qquad j = 1, \ldots, k\,. \\ X \in D^m\,;\, \lambda \geqslant 0.\end{cases} \tag{5}$$

* At first cycle, the feasible region $D^1$ is $D$.




Fig, 1.

The coefficients $\pi^j$ give the relative importance of the distances to the optima, but it must be noted that they are only locally effective and are not of overriding importance as the weights are in the Utility method.

Let us consider the column with index $j$ in the pay-off table $T$. $M^j$ is the maximum value of the column. Let $m^j$ be the minimum value; we choose $\pi^j$ such that

$$\pi^j = \frac{\alpha^j}{\sum_j \alpha^j} \qquad \alpha^j = \frac{M^j - m^j}{M^j} \left( \frac{1}{\sqrt{\sum_{i=1}^{n} (C_i^j)^2}} \right)$$

Term 1 * term 2 **

From term 1, we can make the following assertion: if the value of $C^j$ does not vary much from the optimum solution for varying $X$, the corresponding objective is not sensitive to a variation in the weighting values; so a small weight $\pi^j$ can be assigned to this objective function. As the variation gets larger, the weight $\pi^j$ will become correspondingly bigger. The term 2 normalises the values taken by the objective functions. The $\alpha^j$ are used to define the weights $\pi^j$ in such a way that the sum of $\pi^j$ is 1. This means that different solutions obtained from different weighting strategies can be easily compared.

* This term become $(m^j - M^j)/m^j$ when $M^j \leqslant 0$.
** $C_i^j$ are the coefficients of the objective function $C^j$.

2.4. *Decision phase*

The compromise solution $X^m$ is proposed to the decision-maker, who compares its objective-vector $Z^m$ with $\tilde{Z}$, the ideal one.

If some of the components $C^j(X^m)$ of $Z^m$ are satisfactory and others are not, the decision-maker must accept a certain amount of relaxation of a satisfactory objective $C^j*$ to allow an improvement of the unsatisfactory ones in the next cycle. So he is asked to indicate what $C^j*$ can be relaxed, and the amount of relation $\Delta C^j*$ he can accept

For the next cycle the feasible region is modified

$$D^{m+1} \begin{cases} D^m \\ C^j*(X) \geqslant C^j*(X^m) - \Delta C^j* & (6) \\ C^j(X) \geqslant C^j(X^m), j \neq j^* . & (7) \end{cases}$$

The weight $\pi^j*$ is set to 0 and the calculation phase of cycle $m + 1$ begins. (See the flow-chart in appendix I.)

2.5. *Notes on the algorithm*

1) The analyst can help the decision-maker in determining $j^*$ and the amount of relaxation $\Delta C^j*$ by a standard sensitivity analysis giving the behaviour of the different objective functions in the neighbourhood of the optimum solution $X^m$ (of the problem LP($m$)). This analysis permits, in particular, the following question to be answered:
given two objective functions $C^j$ and $C^{j'}$, what is the upper limit of the increase of $C^{j'}(X)$ when we decide to decrease $C^j(X)$ by unity? (See appendix 2.)
Otherwise, the analyst can help by solving, during the calculation phase of cycle $m$, several LP with feasible regions $D^m$ corresponding to several threshholds $\Delta^j*$ such that $0 < \Delta C_1^j* < \Delta C_2^j* \ldots < \Delta C_M^j*$ .
($\Delta C_M^j*$ is the maximum acceptable relaxation.) In this manner the decision-maker, having seen the different LP ($m$) solutions corresponding to each $\Delta C^j*$, can choose the one he wants, e.g. the one giving the best improvement ratio for the other objectives.

2) Number of iterations. The algorithm produces the solution in fewer cycles than the number $k$ of objective functions, if in the course of the procedure the constraints of type (6) for the $j^*$ and $\Delta C^j*$ chosen will no longer be eligible for further relaxation (the constraints (7) can

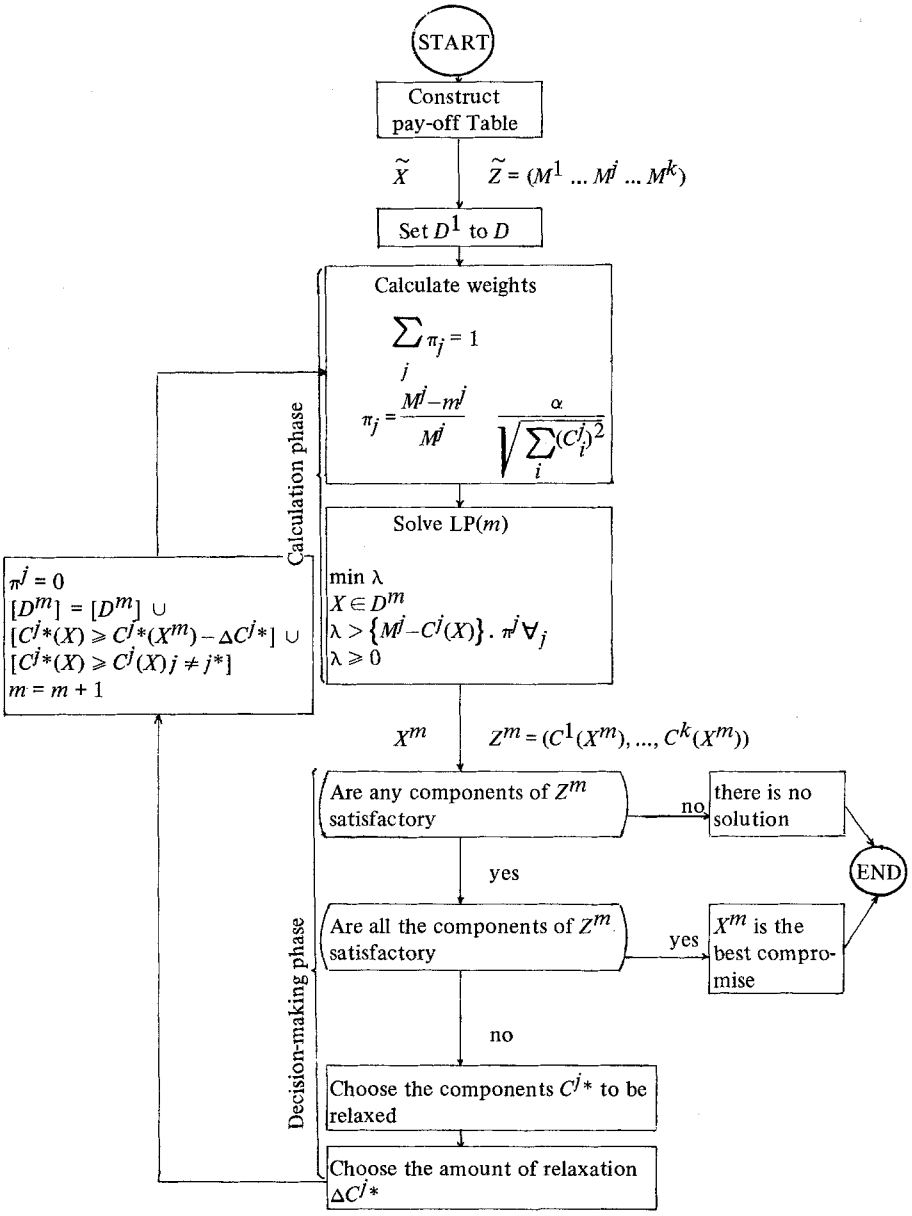
be provisional). Fewer iterations are needed if the decision-maker relaxes several objective functions at once.

3) At the end of the procedure it can be interesting to calculate weights a posteriori, such that:

$$\frac{\pi^j}{M^j - C^j(X^m)} = \text{constant.}$$

4) STEM can be easily programmed for the calculation phase, with a standard LP code which permits successive optimisations of several objective functions. This method has been applied to of Management of Executive Manpower problem [2].

## Appendix 1
Flow-chart of STEM



## Appendix 2

Sensitivity analysis

Let us add to the equations (5) the linear expressions which represent the objectives:

$$\mathrm{LP}(m)\begin{cases}\min z_0 = \lambda \\ z_j = C^j(X) \\ \lambda \geqslant M^j - C^j(X) \quad . \pi^j \\ X \in D^m,\ \lambda \geqslant 0.\end{cases} \quad j = 1, 2, \ldots, k \qquad (8)$$

Let $B^m$ be the optimale basis of (8), $S^m$ be the set of non-basic variables and $P^S$ a column of the tableau such that $s \in S^m$.

Let $Q^s = (B^m)^{-1} . P^s$ be the updated column and $\mu$ be the number of constraints which define $D$.

i.e. $Q^s = (q_0^s, q_1^s, \ldots, q_k^s, \ldots, q_{\mu'}^s)'$ with $\mu' = \mu + 2k$.

Define $S_{\bar{j}}$ as the sub-set of $S^m$ for which $q_j^s < 0$.

Since the $C^j(X)$ are to be maximised, we always will choose $j^*$ such that $S_{\bar{j}} x \neq \emptyset$.

We then can construct a table:

| | $C^1$ | $C^2$ | ... | $C^i$ | ... | $C^k$ |
|---|---|---|---|---|---|---|
| $C^1$ | | | | | | |
| $C^2$ | | | | | | |
| ⋮ | | | | | | |
| $C^j$ | | | | $\delta_{ji}$ | | |
| ⋮ | | | | | | |
| $C^k$ | | | | | | |

such that, if:

$$\alpha)\ \forall s \in S_{\bar{j}} \qquad q_i^s < 0$$

$$\delta_{ji} = \min \qquad q_i^s / q_j^s$$

$$s \in S_{\bar{j}}$$

and $\delta_{ji}$ is a lower bound on the decrease of $C^i(X)$ when we decide to decrease $C^j(X)$ by unity from $X^m$.

$$\beta)\ \exists\, s \in S_{\bar{j}} \qquad \text{such that} \qquad q_i^s \geqslant 0$$

$$\delta_{ji} = \max \qquad -q_i^s / q_j^s$$

$$s \in S_{\bar{j}},\ q_i^s \geq 0$$

and $\delta_{ji}$ is an upper bound on the increase of $C^i(X)$ when we decide to decrease $C^j(X)$ by unity from $X^m$.

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